CSiamese: A Novel Semi-supervised Anomaly Detection Framework for Gas Turbines via Reconstruction Similarity

Abstract

A problem in data-driven anomaly detection of gas turbines is that the monitoring data consists of only a tiny percentage of abnormal samples with the overwhelming majority of normal samples. To address this problem, this paper develops a novel semi-supervised anomaly detection framework, namely CSiamese, and trains on only normal samples, with the ultimate purpose of yielding high detection performance under the condition of unbalanced datasets. First, the convolutional auto-encoder(CAE) is used to learn the reconstructed representation of the input sample with noise. Second, the Siamese network is selected to measure the similarity between the input and corresponding reconstructed representation, which serves as its anomaly score. Besides, a new loss function is developed by improving the contrastive loss, namely triangle loss, and effectively prevents the Siamese network from a collapsed solution using none of the negative sample pairs. Third, maximum likelihood estimation is used to set the proper detection threshold so that the abnormal samples can be distinguished from the normal samples. Finally, the efficacy of the developed CSiamese is demonstrated through the real monitoring data of gas turbines and a public CIFAR-10 data set.

Keywords: anomaly detection, convolutional auto-encoder, the Siamese network, threshold selection

1 Introduction

As a core component of the airplane, gas turbines have a decisive influence on the safety of passengers during a flight. However, due to the harsh work conditions, the minor undetected abnormality will gradually evolve into a catastrophic failure, which will lead to significant economic losses. Therefore, it is necessary to carry out real-time condition monitoring and anomaly detection of gas turbines, and improve the reliability and economy.[1–4] An extremely challenging problem is that monitoring data is highly unbalanced. For example, this paper collects monitoring data of 36 gas turbines in an airline, which

constructs 7593 normal samples and 36 abnormal samples using the sliding window method. It is evident that the total number of abnormal samples is far less than that of normal samples and abnormal samples heavily overlapped with normal samples in the input space, as shown in Figure.1, which undoubtedly increases the difficulty of anomaly detection.

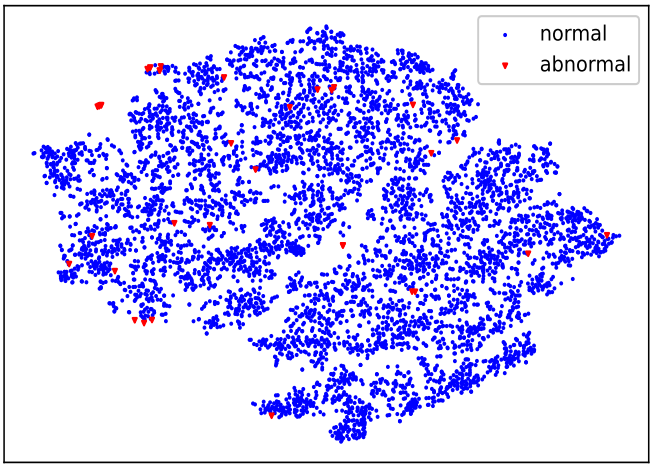


Fig. 1 Two-dimensional visualization of the high-dimensional monitoring data. The red and blue color points respectively represent the abnormal and normal samples.

The anomaly detection methods for gas turbines are usually divided into two categories: model-based[5, 6] and data-driven. The model-based methods establish comprehensive physical models for gas turbines so that anomaly detection results can be explained directly. It is worth noting that the limitations are also obvious. Firstly, they require a large amount of prior knowledge in the modeling process and even some unmeasurable design parameters. Secondly, the modeling process could be very arduous because the damage propagation processes and dynamic responses of complex equipment are very complicated[7]. However, data-driven methods can extract features from real-life monitoring data without understanding of the underlying mechanisms and infer the corresponding health state of the gas turbine[8], which can overcome the abovementioned limitations of the model-based methods. With the development of data mining and cloud computing, data-driven methods have attracted increasing attention in recent years. On one hand, large amounts of monitoring data collected by airlines can not only greatly enhance its availability but also provide more favorable conditions for the implementation of

data-driven methods. On the other hand, many data-driven methods have emerged for different anomaly detection applications[9–12].

As an important branch of the data-driven methods, the most supervised learning methods use both labeled normal and abnormal samples to train a binary classification model for anomaly detection. Firstly, the total number of normal samples is much larger than that of abnormal samples for gas turbines. In this case, the classification model will pay more attention to normal samples during the training phase, which will lead to a sub-optimal diagnostic effect of abnormal samples[13]. As shown in Figure 1, secondly, abnormal samples are quite widely scattered due to different external factors. Thus, abnormal samples collected by airlines are extremely challenging to include all anomaly types. If the abnormality belongs to a new untrained type, the classification model may not even be available to detect it in the test phase[14]. However, the semi-supervised learning methods can effectively alleviate the abovementioned application difficulties of the supervised learning methods on unbalanced datasets. Because the idea of the semi-supervised learning methods is to train the parameters of the model on only normal samples[15]. Recently, a large number of semi-supervised anomaly detection methods have emerged and achieved encouraging performance in the field of anomaly detection, including OC-SVM (one-class support vector machine, OC-SVM)[16, 17], GMM (Gaussian mixture model, GMM)[18], DAE (deep auto-encoder, DAE)[19], SDAE (stacked denoising auto-encoder, SDAE)[20], CAE (convolutional auto-encoder, CAE)[21], and so on. For example, Yan[20] et al. utilized the SDAE to extract features from sensor measurements and then used the one-class extreme learning machine to model normality and detect anomalies. The monitoring data is the two-dimensional time-series data, nevertheless, most anomaly detection methods accept only the one-dimensional vector as input. It is necessary to transform the monitoring data into one-dimensional vectors, which may lose potentially significant information[22]. As an essential variant of the conventional DAE, the CAE contains convolutional layers that are very suitable for the two-dimensional data[23], followed by essential advances in image restoration[24], image clustering[25], and anomaly detection in videos[21, 26]. Therefore, the CAE is worth considering for gas turbine anomaly detection.

Anomaly detection directly using the CAE requires measuring the reconstruction similarity of the sample, that is, the similarity between the input and the corresponding reconstructed representation. However, there are still has two application difficulties. Firstly, measuring the reconstruction similarity depends on the meaningful and computable distance metric in the input space[27]. Researches show that element-wise distance measures are not adequate for complex signal[28, 29]. Secondly, measuring the reconstruction similarity will be inevitably affected by the noise[30]. The selection of the similarity measure is of vital importance. If the inappropriate similarity measure is selected, the measurement effect will be negatively affected[31]. Therefore, it is necessary to explore a general and effective similarity measure. The perfectly natural idea is to learn a mapping $F_W(\cdot)$ from the monitoring data. The raw

data is mapped from the input space to the latent space using the mapping, and a simple distance measure in the latent space, such as Euclidean distance $D_E(\cdot,\cdot)$, can measure the reconstruction similarity of the sample. The study shows that the Siamese network can learn the mapping and resist noise to a certain degree[27, 32], which can solve the abovementioned drawbacks well. Therefore, the Siamese network can be selected to measure the reconstruction similarity of the sample in this paper.

Inspired by the abovementioned studies, we develop a CSiamese model for gas turbine anomaly detection by integrating the CAE with the Siamese network. The core of the developed model is the Siamese network, which is used to measure the reconstruction similarity of the sample by mapping from raw data into a latent space. Moreover, the main contributions of this paper are as follows:

- This paper proposes a novel semi-supervised anomaly detection framework and determines the category of the sample based on its reconstruction similarity, which effectively improves the anomaly detection performance under the condition of unbalanced datasets.
- A triangle loss function is developed by improving the traditional contrastive loss, which can avoid collapsing solutions of the Siamese network even using none of the negative sample pairs.
- The MLETS based on maximum likelihood estimation is proposed to set a proper threshold, which also distinguishes abnormal samples from normal samples in case that the training set only consists of normal samples.
- The experimental results and discussion on real monitoring data of gas turbines and a public CIFAR-10 data set demonstrate the effectiveness of the proposed model against the existing methods.

The rest of this paper is organized as follows. In the first place, Section 2 describes the construction process of the CSiamese, introduces the CAE and the Siamese network, and analyzes the anomaly threshold selection method. Additionally, Section 3 verifies the validity of the proposed model using the real monitoring data of gas turbines and a public CIFAR-10 data set. Finally, Section 4 comes to the main conclusion and identifies the future research work.

2 Material and theory of the developed CSiamese

This section first describes how to construct samples using sliding window method and then demonstrates that the reconstruction similarity of abnormal samples is lower than that of normal samples when they are fed into the CAE that is only trained on normal samples. Additionally, to avoid the negative distance measurement and resist noise to a certain degree, the Siamese network is selected to measure the reconstruction similarity of the sample. Finally, a threshold selection method based on maximum likelihood estimation is designed to accomplish the anomaly detection of gas turbines.

2.1 Data analysis and sample setting

The gas turbine is a highly complex thermal machine composed of fan, low-pressure compressor, high-pressure compressor, combustion chamber, high-pressure turbine, low-pressure turbine, and exhaust system. There is a large number of gas path parameters used for performance monitoring in gas turbines. In this paper, the Delta Exhaust Gas Temperature (DEGT), Delta Fuel Flow (DFF), and Delta Core Speed (DN2), which can best manifest the condition of gas turbines[33], are selected as the abnormal signature parameters (ASP)[34]. Original equipment manufacturers (OEM) also use them for fault diagnosis based on fingerprints.

To train the proposed anomaly detection model, the construction process of the ASS (abnormal signature samples, ASS) is as follows:

(i). The removal date of the gas turbine can be determined according to maintenance reports, and the historical monitoring data of ASP on the wing can be obtained. Therefore, the ASP of the k-th gas turbine are as follows:

$$ASP^{(k)} = \begin{bmatrix} DEGT_1 & DEGT_2 & \cdots & DEGT_{m_k} \\ DN2_1 & DN2_2 & \cdots & DN2_{m_k} \\ DFF_1 & DFF_2 & \cdots & DFF_{m_k} \end{bmatrix} \tag{1}$$

where m_k is the total number of flight cycles selected by the k-th gas turbine.

(ii). The ASS are constructed using sliding window method. According to the relevant CNR (customer notification report, CNR), it is seen that the length of the abnormal sequence is between 5 and 120, and the vast majority is not more than 10. The size of the sliding window is finally set to 10×3, and the step length is set to 5. Additionally, the abnormal sequence is located at the end of the ASP. As a matter of convenience, the sliding window moves from back to front. Therefore, the p-th sample of the k-th gas turbine is as follows:

$$ASS_p^{(k)} = \begin{bmatrix} DEGT_{m_k-5\times p-4} & DEGT_{m_k-5\times p-3} & \cdots & DEGT_{m_k-5\times p+5} \\ DN2_{m_k-5\times p-4} & DN2_{m_k-5\times p-3} & \cdots & DN2_{m_k-5\times p+5} \\ DFF_{m_k-5\times p-4} & DFF_{m_k-5\times p-3} & \cdots & DFF_{m_k-5\times p+5} \end{bmatrix} \tag{2}$$

when $p = 1$, $ASS_p^{(k)}$ is an abnormal sample, and the others are normal samples.

(iii). The set of normal samples is $ASS_n = \{ASS_p^{(k)} : 2 \le p \le n_k, 1 \le k \le N\}$ where n_k is the total number of samples constructed by the k-th gas turbine, and N is the total number of gas turbines. And the set of abnormal samples is $ASS_a = \{ASS_p^{(k)} : p = 1, 1 \le k \le N\}$. Then ASS_n is divided into three groups according to the proportion of 6:2:2. The first 60% of normal samples are the training set, the following 20% are considered as part of the validation set and the rest 20% are part of the test set. Then ASS_a is divided into two groups according to the proportion of 1:1, and 50% and 50% of abnormal samples are respectively regarded as part of the validation and test sets.

2.2 The component analysis of the reconstruction error in the CAE

Like the conventional DAE, the CAE architecture is also composed of an encoder and a decoder. Still, the difference is that the CAE utilizes the convolutional layer and the convolutional transpose layer to replace the fully connected layer. The encoder first maps the input $ASS_p^{(k)}$ to the corresponding latent representation $h_p^{(k)}$, that is $h_p^{(k)} = Encoder(ASS_p^{(k)})$. As an inverse function of the encoder, the decoder maps the latent representation $h_p^{(k)}$ to the reconstructed representation $\widehat{ASS_p^{(k)}}$, that is, $\widehat{ASS_p^{(k)}} = Decoder\left(h_p^{(k)}\right) = Decoder\left(Encoder\left(ASS_p^{(k)}\right)\right) = G\left(ASS_p^{(k)}\right)$. Additionally, the network architecture of the CAE references the structure of the DCGAN[35]. The encoder consists of convolutional layers, batch normalization, and LeakyReLU activation. In contrast, the decoder consists of convolutional transpose layers, batch normalization, and ReLU activation together with a tanh layer at the end. And the architecture schematic diagram of the CAE is shown in Figure.2.

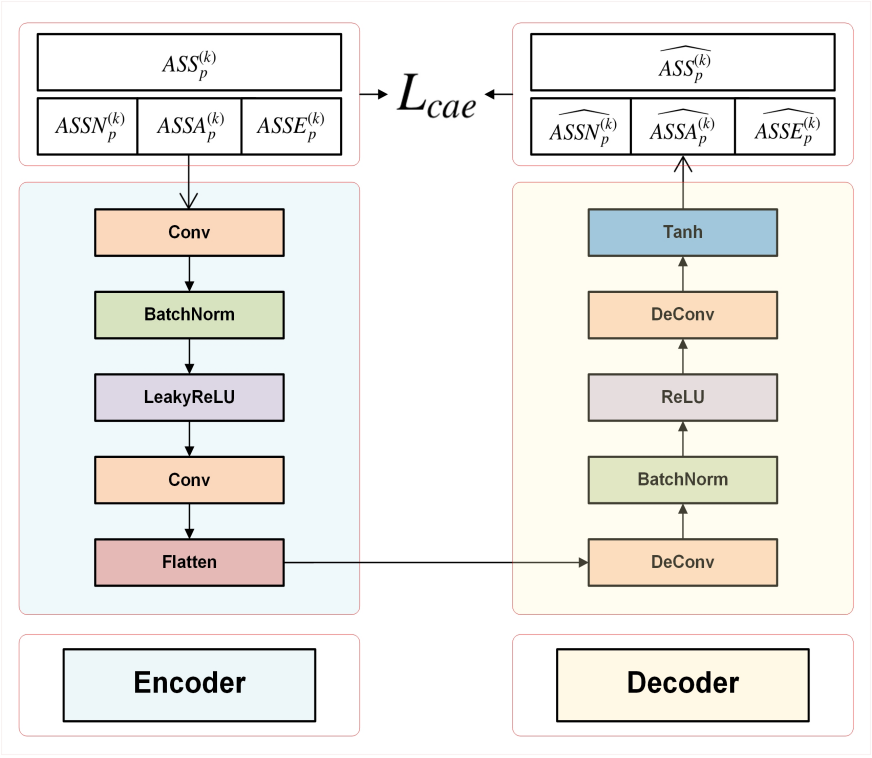


Fig. 2 The architecture schematic diagram of the CAE

The reconstruction error of the CAE is defined as:

$$L_{cae} = \frac{1}{\|ASS\|} \sum_{ASS_p^{(k)} \in ASS} \left\| ASS_p^{(k)} - G\left(ASS_p^{(k)}\right) \right\| \tag{3}$$

where $\|\cdot\|$ is the matrix norm. It is known that using $L_1 - norm$ yields less blurry results than $L_2 - norm$, so $L_1 - norm$ is used to measure the reconstruction loss between $ASS_p^{(k)}$ and $G\left(ASS_p^{(k)}\right)$ in this paper[36]. The parameters can be updated by gradient descent:

$$\bar{d} = \frac{\partial L_{cae}}{\partial G} = \frac{1}{\|ASS\|} \sum_{ASS_p^{(k)} \in ASS} d_p^{(k)} = \frac{1}{\|ASS\|} \sum_{ASS_p^{(k)} \in ASS} \frac{\partial \left\| ASS_p^{(k)} - G\left(ASS_p^{(k)}\right) \right\|}{\partial G} \tag{4}$$

where $d_p^{(k)} = \frac{\partial \left\| ASS_p^{(k)} - G\left(ASS_p^{(k)}\right) \right\|}{\partial G}$ is the contribution of the sample $ASS_p^{(k)}$ to the overall gradient $\bar{d}$. Therefore, the parameter update depends on all the training samples, that is, the CAE tries to reduce the overall error but the error of every single sample[30].

Inspired by the RPCA (Robust Principal Component Analysis, RPCA), the input sample $ASS_p^{(k)}$ is divided into three parts during the training process of the CAE,

$$ASS_p^{(k)} = ASSN_p^{(k)} + ASSA_p^{(k)} + ASSE_p^{(k)} \tag{5}$$

where $ASSN_p^{(k)}$ represents the normal component of the input, $ASSA_p^{(k)}$ represents the abnormal component, and $ASSE_p^{(k)}$ represents the noise component. The CAE is trained only on normal samples. On the one hand, there is no abnormal component $ASSA_p^{(k)}$ in normal samples, so the abnormal component $ASSA_p^{(k)}$ does not contribute to the overall gradient $\bar{d}$. On the other hand, the noise component $ASSE_p^{(k)}$ is included in normal samples but arbitrarily scattered throughout the input space, so its contribution to the overall gradient $\bar{d}$ is offset. To sum up, the overall gradient is dominated by the normal component $ASSN_p^{(k)}$, which means the trained CAE is more likely to reconstruct the normal component $ASSN_p^{(k)}$.

In the test process, the reconstructed representation $\widehat{ASS_p^{(k)}}$ is also divided into three parts:

$$\widehat{ASS_p^{(k)}} = G\left(ASS_p^{(k)}\right) = \widehat{ASSN_p^{(k)}} + \widehat{ASSA_p^{(k)}} + \widehat{ASSE_p^{(k)}} \tag{6}$$

where $\widehat{ASSN_p^{(k)}}$ represents the normal component, $\widehat{ASSA_p^{(k)}}$ represents the

abnormal component, and $\widehat{ASSE_p^{(k)}}$ represents the noise component. Assuming that $D_A(\cdot,\cdot)$ is a meaningful and computable distance metric in the input space, that is, it can properly measure the similarity between the input sample $ASSN_p^{(k)}$ and the reconstructed representation $\widehat{ASSN_p^{(k)}}$, then

(i). $ASSN_p^{(k)} \approx \widehat{ASSN_p^{(k)}} \Rightarrow D_A\left(ASSN_p^{(k)}, \widehat{ASSN_p^{(k)}}\right) \approx 0$

(ii). $D_A\left(ASSA_p^{(k)}, \widehat{ASSA_p^{(k)}}\right) = \begin{cases} 0 & if\ ASS_p^{(k)}\ is\ a\ normal\ sample \\ \rho_p^{(k)} > 0 & if\ ASS_p^{(k)}\ is\ an\ abnormal\ sample \end{cases}$

(iii). $D_A\left(ASSE_p^{(k)}, \widehat{ASSE_p^{(k)}}\right) = \delta_p^{(k)} > 0$

Further, if $ASS_{p_1}^{(k_1)}$ is a normal sample, then $ASSN_{p_1}^{(k_1)} \approx \widehat{ASSN_{p_1}^{(k_1)}}$ and

$$\begin{aligned} D_A\left(ASS_{p_1}^{(k_1)}, \widehat{ASS_{p_1}^{(k_1)}}\right) &= \left\| ASS_{p_1}^{(k_1)} - \widehat{ASS_{p_1}^{(k_1)}} \right\| \\ &= \left\| ASSA_{p_1}^{(k_1)} - \widehat{ASSA_{p_1}^{(k_1)}} + ASSE_{p_1}^{(k_1)} - \widehat{ASSE_{p_1}^{(k_1)}} \right\| \\ &\le D_A\left(ASSA_{p_1}^{(k_1)}, \widehat{ASSA_{p_1}^{(k_1)}}\right) + D_A\left(ASSE_{p_1}^{(k_1)}, \widehat{ASSE_{p_1}^{(k_1)}}\right) \\ &= \delta_{p_1}^{(k_1)} \end{aligned} \tag{7}$$

Thus $0 \le D_A\left(ASS_{p_1}^{(k_1)}, \widehat{ASS_{p_1}^{(k_1)}}\right) \le \delta_{p_1}^{(k_1)}$. If $ASS_{p_2}^{(k_2)}$ is an abnormal sample, then $ASSN_{p_2}^{(k_2)} \approx \widehat{ASSN_{p_2}^{(k_2)}}$

$$\begin{aligned} D_A\left(ASS_{p_2}^{(k_2)}, \widehat{ASS_{p_2}^{(k_2)}}\right) &= \left\| ASSA_{p_2}^{(k_2)} - \widehat{ASSA_{p_2}^{(k_2)}} + ASSE_{p_2}^{(k_2)} - \widehat{ASSE_{p_2}^{(k_2)}} \right\| \\ &\le D_A\left(ASSA_{p_2}^{(k_2)}, \widehat{ASSA_{p_2}^{(k_2)}}\right) + D_A\left(ASSE_{p_2}^{(k_2)}, \widehat{ASSE_{p_2}^{(k_2)}}\right) \\ &= \rho_2^{(k_2)} + \delta_{p_2}^{(k_2)} \end{aligned} \tag{8}$$

$$\begin{aligned} D_A\left(ASS_{p_2}^{(k_2)}, \widehat{ASS_{p_2}^{(k_2)}}\right) &= \left\| ASSA_{p_2}^{(k_2)} - \widehat{ASSA_{p_2}^{(k_2)}} + ASSE_{p_2}^{(k_2)} - \widehat{ASSE_{p_2}^{(k_2)}} \right\| \\ &\ge D_A\left(ASSA_{p_2}^{(k_2)}, \widehat{ASSA_{p_2}^{(k_2)}}\right) - D_A\left(ASSE_{p_2}^{(k_2)}, \widehat{ASSE_{p_2}^{(k_2)}}\right) \\ &= \rho_2^{(k_2)} - \delta_{p_2}^{(k_2)} \end{aligned} \tag{9}$$

Thus $max\left\{0, \rho_2^{(k_2)} - \delta_2^{(k_2)}\right\} \le D_A\left(ASS_{p_2}^{(k_2)}, \widehat{ASS_{p_2}^{(k_2)}}\right) \le \rho_2^{(k_2)} + \delta_{p_2}^{(k_2)}$. In

most cases, the noise component is far less than the abnormal component in abnormal samples, and the reconstruction error of the noise component is also far less than that of the abnormal component, thus $D_A\left(ASS_{p_2}^{(k_2)}, \widehat{ASS_{p_2}^{(k_2)}}\right) > D_A\left(ASS_{p_1}^{(k_1)}, \widehat{ASS_{p_1}^{(k_1)}}\right)$.

Therefore, the reconstruction similarity of the abnormal sample is less than that of the normal sample in most cases, that is, the reconstruction similarity can be used to detect whether the sample is normal or not. However, there are still two drawbacks in the anomaly detection directly using CAE:

(i). Distance Metric: The distance metric $D_A(\cdot,\cdot)$ is vital for calculating the reconstruction similarity of the sample. However, it is not easy to select a proper distance metric for monitoring data, which requires prior knowledge.

(ii). Noise: In extreme cases, the $D_A\left(ASSE_p^{(k)}, \widehat{ASSE_p^{(k)}}\right)$ of the normal sample may be so large that it is wrongly diagnosed as an abnormal sample, so the diagnostic result will also be adversely affected by the noise.

2.3 The improved method of contrastive loss in the Siamese network

The Siamese network has attracted increasing attention in recent years[27, 32, 37, 38]:

(i). LeCun et al.[27] demonstrated that the Siamese network does not require any distance metric in the input space, and a simple distance metric in the latent space, such as Euclidean distance $D_E(\cdot,\cdot)$, can approximate the similarity of the inputs. More precisely, then Euclidean distance $D_E\left(F_W\left(ASS_p^{(k)}\right), F_W\left(\widehat{ASS_p^{(k)}}\right)\right) = \left(\left\|F_W\left(ASS_p^{(k)}\right) - F_W\left(\widehat{ASS_p^{(k)}}\right)\right\|_2\right)^{\frac{1}{2}}$ can measure the similarity of the input $ASS_p^{(k)}$ and its corresponding reconstructed representation $\widehat{ASS_p^{(k)}}$,where $F_W\left(ASS_p^{(k)}\right)$ is the embedded representation of $ASS_p^{(k)}$, and $F_W\left(\widehat{ASS_p^{(k)}}\right)$ is the embedded representation of $\widehat{ASS_p^{(k)}}$.

(ii). Chopra et al.[32] applied the Siamese network to a face verification task, which had a very high degree of variability in the pose, lighting, and position. It can be seen that the Siamese network is not easily disturbed by noise. More precisely, the input $ASS_p^{(k)}$ with noise and the reconstructed representation $\widehat{ASS_p^{(k)}}$ without noise belong to the same category, and thus they will be pulled together[27]. That is, the embedded representation of $ASS_p^{(k)}$ is approximately the same as the embedded representation of $\widehat{ASS_p^{(k)}}$, which weakens

the influence of noise in calculating reconstruction similarity. Therefore, the Siamese network can solve the above two problems well.

2.3.1 The developed triangle loss

Contrastive loss is usually used to train the Siamese network, effectively dealing with paired data[27]. Contrastive loss $L_{sia}\left(W,\left(y,\ ASS,\ \widehat{ASS}\right)\right)$ is defined as below.

$$\begin{aligned} L_{sia} = & \frac{1}{\|ASS\|} \sum_{ASS_p^{(k)} \in ASS}^{K} y_p^{(k)} D_E\left(F_W\left(ASS_p^{(k)}\right), F_W\left(\widehat{ASS_p^{(k)}}\right)\right) \\ & + \left(1 - y_p^{(k)}\right) max\left(d - D_E\left(F_W\left(ASS_p^{(k)}\right), F_W\left(\widehat{ASS_p^{(k)}}\right)\right), 0\right) \end{aligned} \tag{10}$$

where $y_p^{(k)}$ represents the label of paired data $\left\langle ASS_p^{(k)}, \widehat{ASS_p^{(k)}} \right\rangle$, and d is a margin.

However, there are not negative sample pairs in this paper, and the Siamese network may lead to a collapsed solution. More precisely, the contrastive loss becomes

$$L_{sia}\left(W,\left(y,\ ASS, \widehat{ASS}\right)\right) = \frac{1}{\|ASS\|} \sum_{ASS_p^{(k)} \in ASS}^{K} D_E\left(F_W\left(ASS_p^{(k)}\right), F_W\left(\widehat{ASS_p^{(k)}}\right)\right) \tag{11}$$

If all outputs of the Siamese network collapse to a constant, then $D_E\left(F_W\left(ASS_p^{(k)}\right), F_W\left(\widehat{ASS_p^{(k)}}\right)\right) = 0$ and $L_{sia}\left(W,\left(y,\ ASS, \widehat{ASS}\right)\right)$ reaches a minimum. In this case, the Siamese network can not learn a discriminative similarity metric.

To prevent Siamese networks from collapsing, this paper develops the triangle loss by improving the conventional contrastive loss, which is shown in Figure 3.

The triangle loss $\widehat{L}_{sia}\left(W,\left(y, ASS, \widehat{ASS}\right)\right)$ is defined as follows:

$$\widehat{L}_{sia} = \frac{1}{\|ASS\|} \sum_{ASS_p^{(k)} \in ASS}^{K} D_E\left(F_W\left(ASS_p^{(k)}\right), h_p^{(k)}\right) + D_E\left(h_p^{(k)}, F_W\left(\widehat{ASS_p^{(k)}}\right)\right) \tag{12}$$

Thus $0 \leq L_{sia}\left(W,\left(y, ASS, \widehat{ASS}\right)\right) \leq \widehat{L}_{sia}\left(W,\left(y, ASS, \widehat{ASS}\right)\right)$.

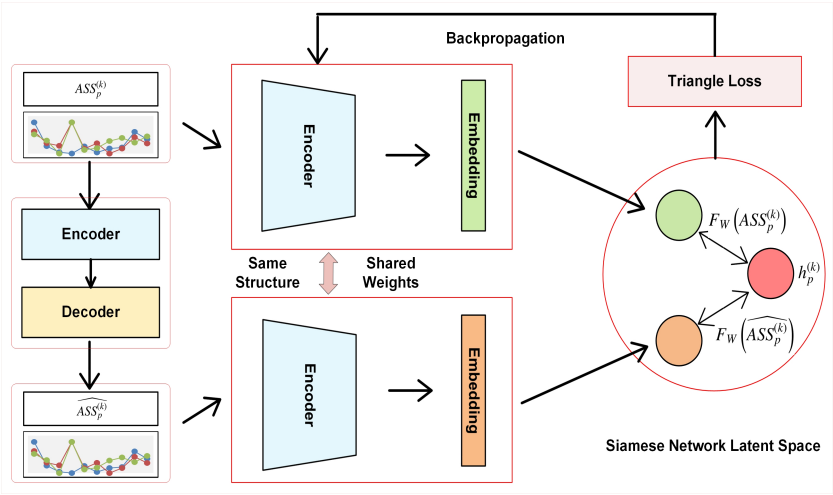


Fig. 3 The developed triangle loss

If $D_E\left(F_W\left(ASS_p^{(k)}\right), h_p^{(k)}\right) \to 0$ and $D_E\left(h_p^{(k)}, F_W\left(\widehat{ASS_p^{(k)}}\right)\right) \to 0$, then $D_E\left(F_W\left(ASS_p^{(k)}\right), F_W\left(\widehat{ASS_p^{(k)}}\right)\right) \to 0$ according to Squeeze Theorem. Obviously, if $\widehat{L}_{sia}\left(W, \left(y, ASS, \widehat{ASS}\right)\right) \to 0$, then $L_{sia}\left(W, \left(y, ASS, \widehat{ASS}\right)\right) \to 0$. Therefore, the input and the reconstructed representation will be pulled together by minimizing $\widehat{L}_{sia}\left(W, \left(y, ASS, \widehat{ASS}\right)\right) \to 0$.

Moreover, if $ASS_{p_1}^{(k_1)} \neq ASS_{p_2}^{(k_2)}$, then $h_{p_1}^{(k_1)} = Encoder\left(ASS_{p_1}^{(k_1)}\right) \neq Encoder\left(ASS_{p_2}^{(k_2)}\right) = h_{p_2}^{(k_2)}$. That is, the label $h_p^{(k)}$ of the paired data $\left\langle ASS_p^{(k)}, \widehat{ASS_p^{(k)}} \right\rangle$ is different for different p and k. The Siamese is trained by optimizing the loss function $\widehat{L}_{sia}\left(W, \left(y, ASS, \widehat{ASS}\right)\right)$. Therefore, all outputs of the Siamese network impossibly collapse to a constant.

2.4 Design of fundamental architectures for CSiamese

The overall framework of the developed CSiamese is shown in Figure 4, which mainly consists of two components: a CAE for obtaining the reconstructed representation of the input and the Siamese network for learning a discriminative similarity metric.

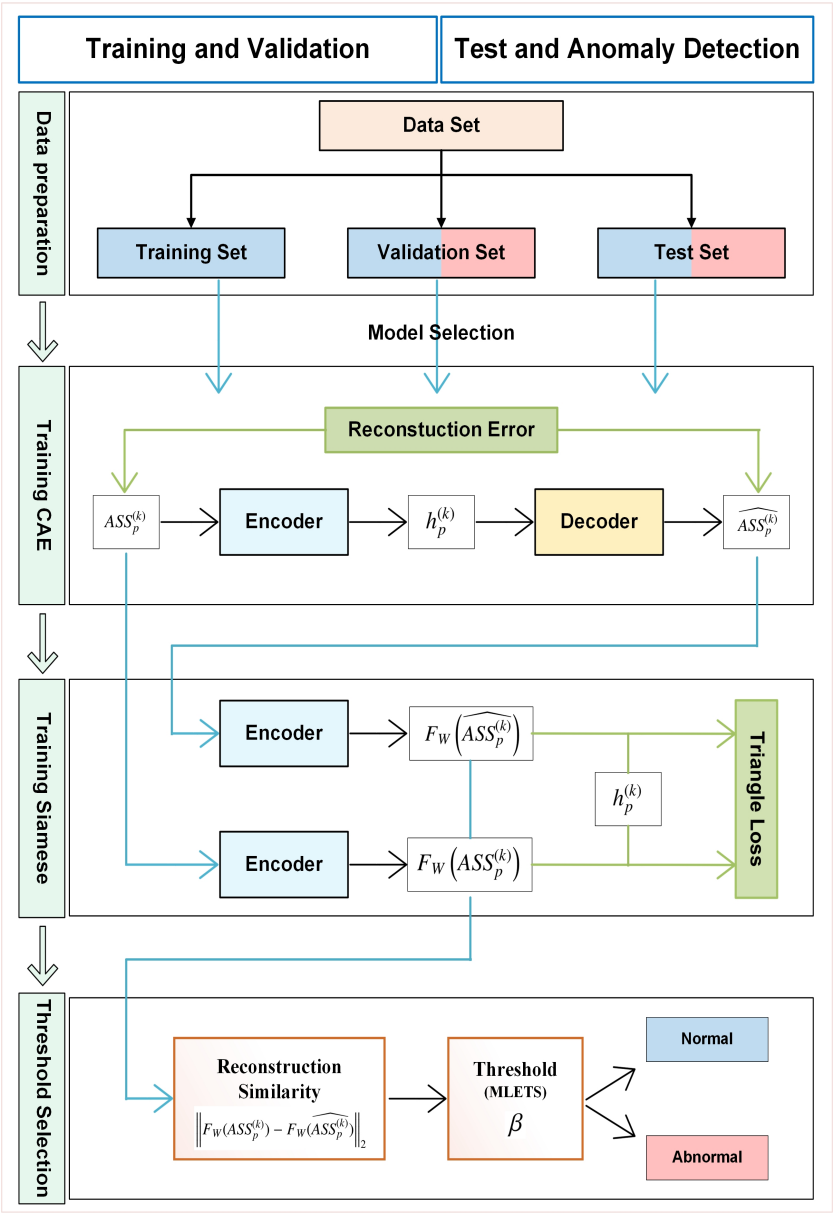


Fig. 4 The overall framework of the developed anomaly detection model

After collecting the monitoring data, it is necessary to train the CSiamese. The training steps are summarized as follows: 1) The CAE is trained using only normal samples, that is, the loss function L_{cae} is minimized by the back-propagation algorithm. 2) The reconstructed representation $\widehat{ASS_p^{(k)}}$

of the input $ASS_p^{(k)}$ is obtained with the well-trained CAE, and the paired data $\left\langle ASS_p^{(k)}, \widehat{ASS_p^{(k)}} \right\rangle$ belongs to the same category. 3) The Siamese network is trained using paired samples, that is, the loss function L_{sia} is minimized by the back-propagation algorithm. Additionally, to ensure that the latent representation of the sample is consistent with the embedded representation in dimensions, the network architecture of the Siamese network is the same as that of the encoder in the CAE. Significantly, there is more than one weight initialization approach, and the common is the random weight initialization. Besides, this paper presents an initialization approach based on weight transfer.

Next, the developed CSiamese is tested using abnormal samples and untrained normal samples. After measuring the reconstruction similarity of the sample using the Siamese network, it is necessary to set the proper threshold so that abnormal samples can be distinguished from normal samples. Therefore, threshold selection is of great importance as well.

2.4.1 Threshold selection based on maximum likelihood estimation

The commonly used threshold selection methods include the empirical method, the histogram method, the Otsu method, and so on[39]. The empirical method requires relevant professional knowledge, and its subjectivity is relatively strong. The histogram method and Otsu method avoid the disadvantages of the empirical method, however, the training set only consists of normal samples in this paper. They are not appropriate according to their calculation process. To address this problem, a new threshold selection method based on maximum likelihood estimation (MLETS) is proposed in this paper.

In this paper, the reconstruction similarity score of the sample is defined as below.

$$s = \left\| F_W(ASS_p^{(k)}) - F_W(\widehat{ASS_p^{(k)}}) \right\|_2 = \sum_{i=1}^{n} \left(F_W^{(i)} \left(ASS_p^{(k)} \right) - F_W^{(i)} \left(\widehat{ASS_p^{(k)}} \right) \right)^2 \tag{13}$$

where $F_W\left(ASS_p^{(k)}\right) = \left(F_W^{(1)}\left(ASS_p^{(k)}\right),\ F_W^{(2)}\left(ASS_p^{(k)}\right),\ \cdots,\ F_W^{(n)}\left(ASS_p^{(k)}\right)\right)$ and $F_W\left(\widehat{ASS_p^{(k)}}\right) = \left(F_W^{(1)}\left(\widehat{ASS_p^{(k)}}\right),\ F_W^{(2)}\left(\widehat{ASS_p^{(k)}}\right),\ \cdots,\ F_W^{(n)}\left(\widehat{ASS_p^{(k)}}\right)\right)$.

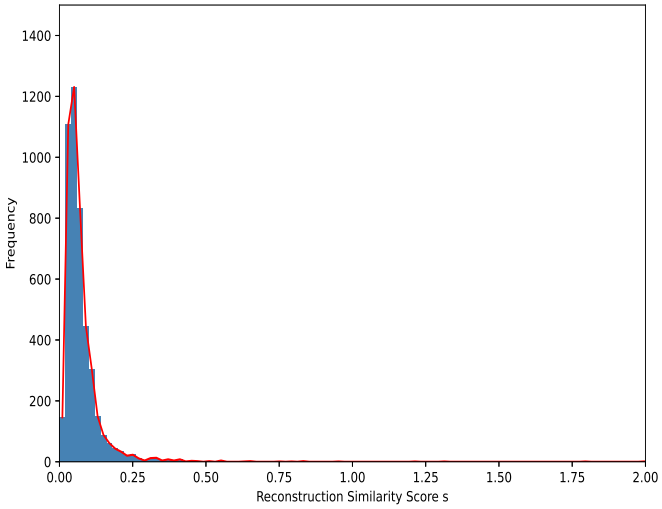


Fig. 5 Frequency histogram of normal samples' reconstruction similarity scores in the training set

Now assume that the reconstruction similarity scores of normal samples in the training set approximately obey the Alpha distribution, which is shown as in Figure 5. The probability density function(PDF) of the reconstruction similarity score s is defined as below.

$$f\left(s;\ \alpha,\mu,c\right)=\frac{1}{\sqrt{2\pi}\Phi\left(\alpha\right)\left(s-\mu\right)^{2}}e^{-\frac{\left(\alpha-\frac{c}{s-\mu}\right)^{2}}{2}} \tag{14}$$

where α, μ, and c are the shape, location, and scale parameter of the Alpha distribution; Φ is the cumulative distribution function(CDF) of the normal distribution. To estimate the parameters α, μ, and c, this can be achieved by maximizing the log-likelihood function of the normal samples in the training set, which is called the maximum likelihood estimate. And the log-likelihood function is defined as below.

$$L(\alpha,\mu,c)=\sum lnf\left(s_{n};\ \alpha,\mu,c\right) \tag{15}$$

Then the maximum likelihood estimation of the parameters α, μ, and c is as follows:

$$\hat{\alpha},\hat{\mu},\hat{c}=\underset{\alpha,\mu,c}{max}\left\{L(\alpha,\mu,c)\right\} \tag{16}$$

According to the research[40], the significance level is set to 0.1, and the confidence interval $[0,\beta]$ can be determined. Finally, the upper limit β of the confidence interval is set as the decision threshold of the CSiamese.

$$r_\beta(s) = \begin{cases} normal, & s < \beta \\ abnormal, & s \geq \beta \end{cases} \tag{17}$$

where $r_\beta(s)$ represents the diagnostic result of the sample. In other words, if the reconstruction similarity score of the sample is less than the decision threshold β, it will be diagnosed as a normal sample. Otherwise, it will be diagnosed as an abnormal sample.

3 Experiments

3.1 Case study I: Gas turbine data set

3.1.1 Data set preparation

In this paper, actual data of gas turbines is collected to evaluate the proposed anomaly detection model, which comes from the real-life operation data of an Asia airline rather than simulated data. Theoretically, some performance parameters deviate from the original trend when gas turbines fail. Among them, DEGT, DN2, and DFF have more direct and obvious changes, which are used for fault diagnosis by the OEM as well. Therefore, DEGE, DN2, and DFF are chosen for anomaly detection. Besides, we collected maintenance reports of the airline from 2008 to 2015. The removal date can be determined according to maintenance reports, and the status of the gas turbine can be judged as well. Then samples are constructed using sliding window method. Additionally, the specific format of the sample is shown in Table 1. Finally, a total of 7629 samples are constructed from the historical monitoring data, as detailed in Table 2.

3.1.2 Experimental setup

Experimentation is implemented in Python 3.7 with Pytorch 1.8.0 using an Intel Core i7-9750H, 2.60GHz processor, and an NVIDIA GeForce GTX1650 GPU. The neural network methods are optimized using Adam with an initial learning rate 0.0002 and a mini-batch size of 64, and momentums $\beta_1 = 0.5$, $\beta_2 = 0.999$[41]. We respectively train the model for 4, 6 epochs for the CAE and Siamese according to trial tests. Due to length limitations, the choice of the architecture hyperparameters is clearly introduced in Appendix.

3.1.3 Evaluation metrics

If the number of abnormal samples is close to that of normal samples, the diagnostic performance of the anomaly detection model can be evaluated well using the overall accuracy. However, the number of normal samples is much larger than that of abnormal samples for gas turbines, as detailed in Table 2. The overall accuracy has no vital reference value in this case. For example, the number of normal samples is 1524, while that of abnormal samples is only 18 in the test set. Even if all the test samples are diagnosed as normal samples,

Table 1 The specific sample format of gas turbine data set

No	DEGT	DN2	DFF	State
1	30.6606	-0.2946	0.6565	nomal
	22.4841	-0.6491	0.234	
	20.3759	-0.7301	-0.657	
	19.9201	-0.0157	1.4827	
	22.3744	-0.8221	-0.408	
	20.2971	-0.6535	-0.2862	
	21.6608	-0.9676	0.2019	
	21.9958	-0.814	0.415	
	27.3981	-0.471	0.0942	
	24.8495	-0.6588	0.5205	
	...	...	...	
7629	28.0694	-0.9336	2.3797	abnomal
	23.4776	-0.8427	1.4983	
	13.1551	-1.2589	-0.0978	
	39.0245	-0.8254	4.3871	
	35.0468	-0.716	2.1861	
	17.7177	-1.1165	0.7452	
	16.949	-0.9617	0.38	
	16.2494	-1.194	-0.6943	
	22.766	-1.0596	2.0265	
	15.8263	-1.1538	-0.0533	

Table 2 Detail description of gas turbine data set

State	Training	Validation	Test	Total
Normal	4544	1525	1524	7593
Abnormal	0	18	18	36
Total	4544	1543	1542	7629

the overall accuracy is still as high as 98.83%. Obviously, the overall accuracy is not suitable for the unbalanced dataset. The AUROC and confusion matrix are selected to evaluate the proposed anomaly detection model in this paper.

AUROC: The area under the ROC (receiver operating characteristic, ROC) curve is defined as AUROC. The study shows that AUROC can better measure the performance of the classifier than the overall accuracy under the condition of unbalanced data[42].

Confusion Matrix: The diagnostic efficacy of normal samples can be evaluated by true negative rate (TNR), while the diagnostic efficacy of abnormal samples can be evaluated by the true positive rate (TPR). They can be easily calculated according to the confusion matrix (Table 3). Since the above-mentioned evaluation criteria are independent, they will not be disturbed by the unbalanced dataset.

$$TPR = \frac{TP}{TP + FN} \tag{18}$$

Table 3 An illustration of confusion matrix

Actual Class	**Predicted Class**	
	Positive	**Negative**
Positive	True Positive (TP)	False Negative (FN)
Negative	False Positive (FP)	True Negative (TN)

$$TNR = \frac{TN}{TN + FP} \tag{19}$$

3.1.4 Experimental results and discussion

3.1.4.1 The effectiveness of the proposed anomaly detection method

To verify the effectiveness of the proposed CSiamese, this paper establishes five groups of comparison experiments. In the first group of experiments, the OC-SVM is directly used to diagnose the raw data. The selected kernel types include the common linear, polynomial, radial basis function, and sigmoid kernel functions, which are separately recorded as the OC-SVM(Linear), OC-SVM(Ploy), OC-SVM(RBF), and OC-SVM(Sigmoid). The parameter nu, an upper bound on the fraction of margin errors and a lower bound of the fraction of support vectors relative to the total number of training examples, is set to be 0.001, 0.0001, 0.01, and 0.0001. In the second group of experiments, the raw data is directly classified by the GMM. The selected types of covariance parameters include the common full, tied, diag, spherical, which are separately recorded as the GMM(Full), GMM(Tied), GMM(Diag), and GMM(Spherical). The number of mixture components is set to be 512, 96, 4, and 4. In the third group of experiments, the conventional DAE with five layers is directly used to diagnose the raw data. The input layer has thirty nodes, while the output layer has thirty nodes. On the one hand, the number of nodes in the hidden layers is challenging to select. On the other hand, the comparison experiment shows the potential of the CSiamese rather than conducting a comprehensive performance comparison. Therefore, the number of nodes in the hidden layers is determined after several initial experiments in this paper. They respectively have 110, 55, 55, and 110 nodes. In the fourth group of experiments, the CAE is directly used to detect the raw data, and the network architectures of the encoder and decoder are the same as that of the encoder and decoder in the CSiamese. The channels of the first convolution layer are set to 32, and that of the second convolution layer are set to 64. Significantly, there is more than one weight initialization approach for the Siamese network. The common is the random weight initialization, and the other is an initialization approach based on weight transfer. In the fifth group of experiments, the CAE is used to obtain the paired samples, and then the paired samples are diagnosed using the Siamese network. The network architecture of the Siamese network is consistent with the encoder of the CAE, and the weights are initialized randomly. The proposed CSiamese is similar to the fifth, and the only difference is that the

weights of the Siamese network are transferred from the well-trained encoder. As a matter of convenience, the fifth can be recorded as CSiamese(RW). The hyperparameter setup of the comparison experiments is given in Appendix as well.

Table 4 AUROC(%) and inference time(milliseconds/batch) of the different anomaly detection methods on the gas turbine data set

No.	Method	AUROC	Inference Time
1st	OC-SVM(Linear)	60.48	0.0889
	OC-SVM(Poly)	59.33	**0.0870**
	OC-SVM(RBF)	92.38	0.5706
	OC-SVM(Sigmoid)	60.04	0.0985
2nd	GMM(Full)	16.35	13.0035
	GMM(Tied)	8.16	2.2331
	GMM(Diag)	14.21	0.1468
	GMM(Spherical)	13.78	0.1680
3rd	DAE	91.69	0.3147
4th	CAE	96.62	1.2451
5th	CSiamese(RW)	97.16	1.8173
Proposed method	CSiamese	**97.53**	1.8382

Table 4 shows the AUROC of each group of experiments. The following conclusions can be drawn. Compared with the conventional machine learning methods such as OC-SVM, GMM, and DAE, first of all, the CAE achieves the best diagnostic results. More specifically, compared with the OC-SVM(Linear), OC-SVM(Poly), OC-SVM(RBF), OC-SVM(Sigmoid), GMM(Full), GMM(Tied), GMM(Diag), GMM(Spherical), and DAE, the AUROC of the CAE in the test set is improved by about 36.14%, 37.29%, 4.24%, 36.58%, 80.27%, 88.46%, 82.41%, 82.84%, and 4.93%. Undoubtedly, the performance advantage of the CAE benefits from the convolutional layer and convolutional transpose layer. That is, the convolutional layer and convolutional transpose layer can capture more latent and significant information from the two-dimensional data, which is very suitable for the monitoring data of gas turbines. Secondly, the proposed CSiamese obtains the optimal detection performance among all the methods used in the experiments, and the CSiamese(RW) comes second as well. The AUROC of the CSiamese(RW) and CSiamese is improved by about 0.54% and 0.91% than that of the CAE. It is evident that the Siamese network can better measure the reconstruction similarity of the sample and thus better distinguish abnormal samples from normal samples. Thirdly, the AUROC of the CSiamese(RW) is smaller than that of the CSiamese. It can be seen that the weight transfer can improve the detection performance of the model.

Table 4 also shows the average batch inference time of the different anomaly detection models during the testing period. It is easy to observe that the OC-SVM(Poly) has the fastest inference time and takes about 0.0870 milliseconds to detect sixty-four test samples, while the proposed method takes over 1.8382 milliseconds. The difference between the CSiamese and OC-SVM(Poly) is about 1.7512 milliseconds, which can be acceptable in engineering practice. Additionally, compared with the OC-SVM(Poly), the AUROC of the CSiamese is improved by about 38.2%, which significantly improves the anomaly detection performance. On the basis of comprehensive consideration, the proposed CSiamese obtains the optimal detection performance, which is the most potential for gas turbine anomaly detection in engineering practice.

3.1.4.2 The effectiveness of the proposed threshold selection method

The effectiveness of the proposed MLETS is discussed in this subsection. First of all, the maximum likelihood estimation of the parameters α, μ, and c is 3.3068, -0.0529, and 0.3646. Then, the 90% confidence interval is $[0, 0.1270]$. Finally, the decision threshold β is set to 0.1270. The reconstruction similarity scores of most normal samples and very few abnormal samples in the test set are less than the threshold β. Thus, it's very effective to diagnose whether the sample is normal or not.

Table 5 Confusion matrix of the proposed threshold selection method

Data set	Actual Class	Predicted Class	
		Abnormal	**Normal**
Validation	**Abnormal**	**16**	2
	Normal	149	**1375**
Test	**Abnormal**	**16**	2
	Normal	159	**1364**

Table 5 shows the diagnostic results of the proposed MLETS. In the validation set, there are 1524 normal samples, and 1375 normal samples are correctly diagnosed, so the true negative rate is about 90.22%. Besides, 16 abnormal samples are correctly diagnosed from 18 abnormal samples, and the true positive rate is about 88.89%. The true negative rate of the test set is 89.56%, and the true positive rate is also about 88.89%. The results show that the proposed threshold selection method can effectively distinguish abnormal samples from normal samples in case that the training set only consists of normal samples.

3.2 Case study II: CIFAR-10 data set

3.2.1 Data set preparation

To further validate the effectiveness and superiority of the proposed CSiamese, the CIFAR-10 data set is also selected to evaluate the anomaly detection performance. This data set consists of 60000 images in ten classes, with 6000

images per class. The splitting strategy for the data set is as follows. First of all, we treat one class as the abnormal class, while the union of the rest classes is considered as the normal class. In total, there are ten situations, each of which consists of 54000 normal samples and 6000 abnormal samples. Secondly, normal samples are divided into three groups according to the proportion of 8:1:1, and 80%, 10%, and 10% of normal samples are respectively regarded as part of the training, validation, and test sets. Unlike normal samples, abnormal samples are divided into two groups according to the proportion of 1:1, and 50% and 50% of abnormal samples are respectively considered as part of the validation and test sets. That is, there are no abnormal samples in the training set. Finally, the splitting for the data set is detailed in Table 6.

Table 6 Detail description of CIFAR-10 data set

State	Training	Validation	Test	Total
Normal	43200	5400	5400	54000
Abnormal	0	3000	3000	6000
Total	43200	8400	8400	60000

3.2.2 Experimental results and discussion

In this subsection, the detection performance of the proposed CSiamese is only compared with the DAE, CAE, and CSiamese(RW). That is, comparison experiments do not include the OC-SVM and GMM due to their poor performances. Furthermore, to verify the anti-noise performance of the Siamese network in measuring the reconstruction similarity of the sample, the AWGN (Additive White Gaussian Noise, AWGN) provided by the module skimage exposure is selected to be added to the raw data. It is worth noting that the raw data will be submerged by noise if the noise intensity is very high, while the raw data will hardly be affected by noise if the noise intensity is very low. Either case can not meet the purpose of the experiment. Thus, the mean of the AWGN is its default value 0, and the variance of the AWGN is set to four noise levels (i.e., 0.01, 0.0625, 0.25, 1).

Table 7 shows the area under the ROC curve of the different anomaly detection models on the CIFAR-10 data set. It can be seen that the proposed CSiamese model outperforms all other methods on seven of the ten situations and is also competitive on the remaining one. This once again verifies the effectiveness and superiority of the proposed CSiamese. Nevertheless, there are two situations where CSiamese does not perform well, as shown in Figure 6. For example, the CSiamese performs badly on the situation that the car is the abnormal class. It is not difficult to discover that the car is visually similar to several other classes like ship and truck. The CAE in the framework is well trained on the training set containing the ship and truck, which can produce a relatively good reconstruction for the car. In addition, the Siamese network in

Table 7 AUROC(%) and ranking of the different anomaly detection methods on the CIFAR-10 data set

Abnomal Class	**Methods**			
	DAE	**CAE**	**CSiamese(RW)**	**CSiamese**
plane	42.05	96.16	98.12	**98.48**
car	73.31	**86.82**	66.81	67.97
bird	38.3	54.78	63.31	**66.59**
cat	56.74	**64.52**	62.62	62.84
deer	33.94	76.19	83.6	**85.98**
dog	58.89	65.33	77.82	**79.43**
frog	41.86	91.04	94.17	**95.19**
horse	62.79	69.44	72.75	**74.89**
ship	40.22	86.78	95.13	**96.15**
truck	74.35	**86.75**	68.98	73.15
Average AUROC	52.25	77.78	78.33	**80.07**

the framework pulls the input and its corresponding reconstructed representation together, thus weakening the difference between normal and abnormal samples as well. That is, if abnormal samples are similar to partial normal samples, the proposed model will face important challenges. Therefore, our future efforts will concentrate on the discovery of a more effective anomaly detection method that can consider the diversity of normal components explicitly and weaken the representation capacity of abnormal components.

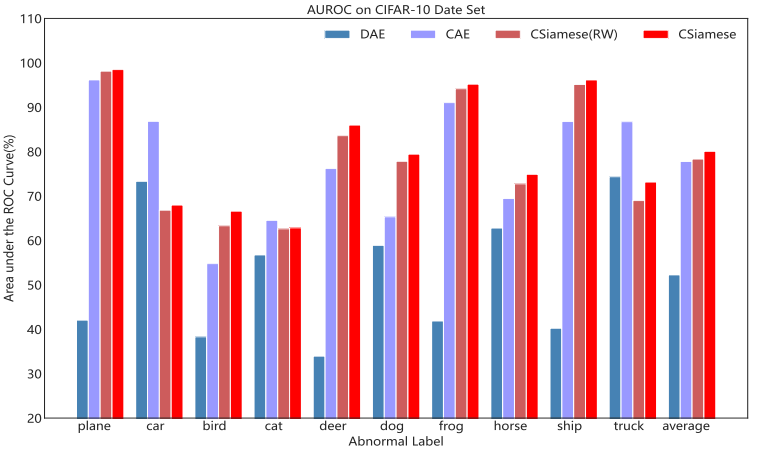


Fig. 6 Area under the ROC curve for the CIFAR-10 data set

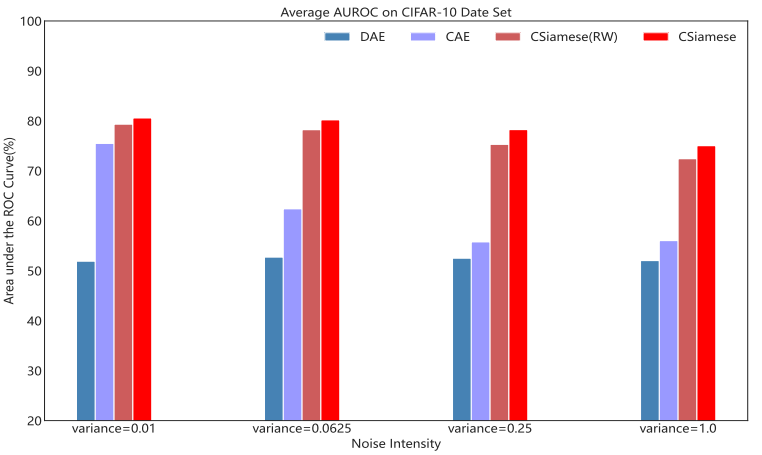


Fig. 7 Average AUROC of the different anomaly detection methods in different noise intensity conditions

Table 8 shows the AUROC of the DAE, CAE, CSiamese(RW), and CSiamese in different noise intensity conditions. The experimental results show that the diagnostic accuracy of all models except the DAE decreases under the condition of noise with different intensities. The higher the noise intensity, the worse the diagnostic accuracy. The diagnostic accuracy of CSiamese(RW) and CSiamese is better than that of the DAE and CAE in the same noise intensity conditions, which indicates that the Siamese network has good anti-noise performance in measuring the reconstruction similarity of the sample. The CAE directly takes the reconstruction error as the reconstruction similarity of the sample, which is easy to interfere by noise. However, the Siamese network pulls the input and its corresponding reconstructed representation together, thus effectively weakening the influence of noise. Furthermore, Figure 7 intuitively represents the trend of the diagnostic accuracy for the above models in different noise intensity conditions.

4 Conclusions

This paper develops a novel anomaly detection framework, namely CSiamese, which can alleviate the problem of the unbalanced dataset. The CAE is trained to reconstruct the normal component of the input at first, and then the Siamese network is used to measure the similarity between the input with noise and the corresponding reconstructed representation. Besides, the MLETS is used to set the proper threshold so that abnormal samples can be distinguished from normal samples.

The effectiveness of the proposed CSiamese architecture is verified through the real monitoring data of gas turbines and a public CIFAR-10 data set. For

Table 8 AUROC(%) and ranking of the different anomaly detection methods in different noise intensity conditions

Noise	Abnomal	Methods			
Intensity	**Class**	**DAE**	**CAE**	**CSiamese(RW)**	**CSiamese**
var=0.01	plane	40.33	97.29	**98.59**	98.22
	car	73.16	**77.94**	68.06	71.05
	bird	38.9	57.4	63.4	**66.14**
	cat	56.11	57.68	63.06	**69.36**
	deer	34.81	76.8	86.67	**86.77**
	dog	58.93	63.63	79.3	**81.25**
	frog	41.7	91.11	94.55	**97.77**
	horse	62.94	66.76	**78.01**	70.24
	ship	39.47	88.89	95.11	**96.41**
	truck	72.56	**77.19**	66.44	68.44
	Average	51.89	75.47	79.32	**80.57**
var=0.0625	plane	41.84	80.33	96.73	**97.18**
	car	**65.32**	50.54	62.7	64.85
	bird	46.61	63.43	**65.04**	62.4
	cat	53.06	48.81	67.71	**68.28**
	deer	44.44	75.1	86.71	**89.82**
	dog	60.64	51.39	78.5	**82.19**
	frog	44.88	75.79	**96.49**	95.89
	horse	62.92	57.29	67.01	**74.73**
	ship	41.86	70.83	**94.2**	93.12
	truck	65.55	50.27	67.12	**73.43**
	Average	52.71	62.38	78.22	**80.19**
var=0.25	plane	45.29	68.28	89.39	**99.21**
	car	54.16	41.46	65.26	**73.49**
	bird	56.07	61.36	59.89	**63.6**
	cat	50.41	47.25	68.21	**69.39**
	deer	56.64	70.05	**83.08**	82.08
	dog	56.53	50.1	**75.42**	73.63
	frog	50.56	66.02	**96.25**	93.08
	horse	58.32	52.44	68.85	**72.5**
	ship	45.47	60.44	85.58	**94.87**
	truck	51.4	40.23	**60.9**	60.56
	Average	52.49	55.76	75.28	**78.24**
var=1.0	plane	46.66	70.08	**93.06**	92.65
	car	49.74	42.9	58.36	**65.66**
	bird	57.44	59.76	**64.15**	63.25
	cat	50.65	49.07	62.98	**68.03**
	deer	58.16	66.29	80.97	**83.28**
	dog	54.69	50.79	**71.73**	71.72
	frog	52.23	64.58	83.26	**87.51**
	horse	56.13	51.7	70.13	**73.72**
	ship	46.85	63.43	**82.98**	81.61
	truck	47.61	41.59	56.46	**62.77**
	Average	52.02	56.02	72.41	**75.02**

gas turbine data set, compared with the OC-SVM(Linear), OC-SVM(Poly),

OC-SVM(RBF), OC-SVM(Sigmoid), GMM(Full), GMM(Tied), GMM(Diag), GMM(Spherical), DAE, and CAE, the AUROC of the CSiamese in the test set is improved by about 37.05%, 38.20%, 5.15%, 37.49%, 81.18%, 89.37%, 83.32%, 83.75%, 5.84%, and 0.91%. The CSiamese model also outperforms all other methods on seven of the ten situations and is also competitive on the remaining one on the CIFAR-10 data set. The reason is that the convolutional layers in the model are suitable to deal with two-dimensional data, and the Siamese network in the model is suitable to measure the reconstruction similarity of the sample even under the influence of noise. Moreover, the MLETS is also investigated experimentally and tested to be effective.

Finally, the anomaly detection of gas turbines is a developing research topic. And our future efforts will concentrate on the discovery of a more effective anomaly detection method that can consider the diversity of normal components explicitly and weaken the representation capacity of abnormal components.

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Appendix A Gas Turbine Data Set Experimental Details

A.1 Architecture and hyperparameters of the OC-SVM

Table A1 shows how the choice of the hyperparameter nu specifically affect the AUROC(%) and ranking of the OC-SVM model with different kernel functions. It can be seen that the optimal ranking of the OC-SVM with linear kernel is achieved when the nu is 0.001 for the gas turbine data set. The rest respectively are 0.0001, 0,01, and 0.0001, which is shown in Table A2.

Table A1 AUROC(%) and ranking of the OC-SVM on the validation set

Method	Nu	AUROC	Ranking	Method	Nu	AUROC	Ranking
	0.0001	53.88	2		0.0001	93.12	4
	0.001	53.95	**1**		0.001	92.78	6
	0.005	51.02	3		0.005	93.22	3
	0.01	50.29	4		0.01	93.6	**1**
	0.05	48.08	10		0.05	93.32	2
	0.1	47.73	12		0.1	92.94	5
OC-SVM	0.15	47.49	14	OC-SVM	0.15	92.5	7
(Linear)	0.2	47.69	13	(RBF)	0.2	91.89	9
	0.25	47.88	11		0.25	91.89	9
	0.3	48.2	9		0.3	91.88	10
	0.35	48.72	8		0.35	91.6	11
	0.4	49.32	7		0.4	91.56	12
	0.45	49.76	6		0.45	91.4	13
	0.5	50.15	5		0.5	91.33	14
	0.0001	53.86	**1**		0.0001	54.63	**1**
	0.001	53.62	2		0.001	54.19	2
	0.005	49.19	5		0.005	52.83	3
	0.01	49.45	3		0.01	52.33	4
	0.05	47.11	10		0.05	51.06	9
	0.1	46.75	13		0.1	50.93	11
OC-SVM	0.15	46.64	14	OC-SVM	0.15	50.85	14
(Poly)	0.2	46.84	12	(Sigmoid)	0.2	50.85	14
	0.25	47.05	11		0.25	50.91	12
	0.3	47.45	9		0.3	51.02	10
	0.35	47.93	8		0.35	51.11	8
	0.4	48.4	7		0.4	51.21	7
	0.45	48.83	6		0.45	51.25	6
	0.5	49.25	4		0.5	51.3	5

Table A2 Architecture and hyperparameters of the OC-SVM

Operation	Kernel	Nu
OC-SVM	Linear	0.001
	Poly	0.0001
	RBF	0.01
	Sigmoid	0.0001

A.2 Architecture and hyperparameters of the GMM

Table A3 shows how the choice of the hyperparameter mixture components(mc) specifically affect the AUROC(%) and ranking of the GMM model with different covariance types. It can be seen that the optimal ranking of the GMM with full covariance type is achieved when the number of mixture components is 512 for the gas turbine data set. The rest respectively are 96, 4, and 4, which is shown in Table A4.

Table A3 AUROC(%) and ranking of the GMM on the validation set

Method	Mc	AUROC	Ranking	Method	Mc	AUROC	Ranking
GMM (Full)	4	7.17	2	GMM (Diag)	4	11.89	**1**
	8	6.12	4		8	6.47	2
	16	6.1	5		16	5.23	3
	32	5.76	6		32	5.01	4
	64	5.61	7		64	3.8	5
	96	4.48	9		96	3.77	6
	128	4.32	10		128	3.42	7
	192	7.06	3		192	2.58	9
	256	5.17	8		256	2.41	10
	512	14.28	**1**		512	3.11	8
GMM (Tied)	4	6.54	6	GMM (Spherical)	4	12.43	**1**
	8	6.54	6		8	8.4	2
	16	6.26	9		16	4.52	3
	32	6.34	7		32	3.22	6
	64	6.72	3		64	1.87	10
	96	7.73	**1**		96	1.99	8
	128	6.56	4		128	3.24	5
	192	7.13	2		192	3.33	4
	256	6.28	8		256	2.6	7
	512	5.14	10		512	1.87	10

A.3 Architecture and hyperparameters of the DAE

Table A5 shows how the choice of the hyperparameter specifically affect the AUROC(%) and ranking of the DAE model. It can be seen that the optimal ranking of the DAE is achieved when the dimension of the latent representation is 55 for the gas turbine data set, which is shown in Table A6.

Table A4 Architecture and hyperparameters of the GMM

Operation	Covariance Type	Mc
GMM	Full	512
	Tied	94
	Diag	4
	Spherical	4

Table A5 AUROC(%) and ranking of the DAE on the validation set

Method	Latent Dimension	AUROC	Ranking
DAE	5	0.8375	8
	10	0.8303	9
	15	0.822	10
	20	0.8157	11
	25	0.8146	12
	30	0.8426	7
	35	0.9618	4
	40	0.962	2
	45	0.9614	6
	50	0.9617	5
	55	0.9646	**1**
	60	0.9619	3

Table A6 Architecture and hyperparameters of the DAE

Operation	Units	Non Linearity	Dropout
Encoder			
Dense	110	Tanh	0.0
Dense	55	Tanh	0.0
Decoder			
Dense	55	Tanh	0.0
Dense	110	Tanh	0.0
Optimizer	Adam		
Batch size	64		
Latent dimension	55		
Epochs	10		
Weight initialization	Gaussian(μ=0, σ=0.02)		
Bias initialization	Constant(0)		

A.4 Architecture and hyperparameters of the CAE

Table A7 shows how the choice of the hyperparameter specifically affect the AUROC(%) and ranking of the CAE model. It can be seen that the optimal ranking of the CAE is achieved when the dimension of the latent representation is 128 for the gas turbine data set, which is shown in Table A8.

Table A7 AUROC(%) and ranking of the CAE on the validation set

Method	Latent Dimension	AUROC	Ranking
CAE	8	0.756	7
	16	0.8616	6
	32	0.9277	5
	64	0.9693	3
	128	0.9814	**1**
	256	0.9767	2
	512	0.9487	4

Table A8 Architecture and hyperparameters of the CAE

Operation	Kernel	Strides	Filters Units	Non Linearity	Batch Norm
Encoder					
Conv	4×4	1×2	32	LeakyReLU(0.2)	✓
Conv	4×4	1×2	64	None	×
Decoder					
DeConv	4×4	1×2	64	ReLU	✓
DeConv	4×4	1×2	32	Tanh	×

Optimizer	Adam
Batch size	64
Latent dimension	128
Epochs	10
Weight initialization	Gaussian(μ=0, σ=0.02)
Bias initialization	Constant(0)

A.5 Architecture and hyperparameters of the CSiamese

Table A9 shows how the choice of the hyperparameter specifically affect the AUROC(%) and ranking of the CSiamese model. It can be seen that the optimal ranking of the CSiamese is achieved when the dimension of the latent representation is 128 for the gas turbine data set, which is shown in Table **??**.

Table A9 AUROC(%) and ranking of the CSiamese on the validation set

Method	Latent Dimension	AUROC	Ranking
CSiamese	8	0.8879	6
	16	0.835	7
	32	0.9005	5
	64	0.9299	4
	128	0.9686	**1**
	256	0.9619	2
	512	0.9545	3

Table A10 Architecture and hyperparameters of the CSiamese

Operation	Kernel	Strides	Filters Units	Non Linearity	Batch Norm
Encoder					
Conv	4×4	1×2	32	LeakyReLU(0.2)	✓
Conv	4×4	1×2	64	None	×
Decoder					
DeConv	4×4	1×2	64	ReLU	✓
DeConv	4×4	1×2	32	Tanh	×
Siamese					
Conv	4×4	1×2	32	LeakyReLU(0.2)	✓
Conv	4×4	1×2	64	None	×
Optimizer	Adam				
Batch size	64				
Latent dimension	128				
Epochs	(4, 6)				

Appendix B CIFAR-10 Experimental Details

B.1 Architecture and hyperparameters of the DAE

Table B11 and Table B12 depict how the choice of the hyperparameter specifically affect the AUROC(%) and ranking of the DAE model. It can be seen that the optimal ranking of the DAE is achieved when the dimension of the latent representation is 12 for the gas turbine data set, which is shown in Table B13.

Table B11 AUROC(%)/Ranking of the DAE on the validation set

Abnomal Class	Latent Dimension						
	2	4	6	8	10	12	16
plane	46.71/3	38/13	39.74/10	39.08/12	39.53/11	41.44/6	40.06/8
car	67.08/13	70.58/12	73.52/3	72.16/11	72.38/10	72.95/7	72.94/8
bird	36.1/13	37.14/11	37.28/8	37.15/10	37.3/7	38.24/5	37.17/9
cat	56.47/2	58.66/1	56.12/4	56.13/3	55.64/6	55.55/7	55.92/5
deer	31.13/13	32.16/12	32.64/10	33.14/9	33.61/7	33.83/6	32.61/11
dog	59.7/2	60.28/1	59.33/3	58.02/4	57.35/6	57.75/5	57.15/7
frog	38.21/13	38.63/12	39.21/11	41.58/10	41.84/9	41.85/8	43.37/7
horse	58.09/13	60.4/12	61.93/4	61.73/7	62.78/3	63.24/1	62.98/2
ship	50.87/1	43.15/3	43.19/2	41.5/5	40.35/9	41.45/6	40.59/8
truck	70.22/13	72.79/10	74.52/2	73.95/5	73.42/8	74.42/3	74.61/1
Average	8.6	8.7	5.7	7.6	7.6	**5.4**	6.6

B.2 Architecture and hyperparameters of the CAE

Table B14 and Table B15 depict how the choice of the hyperparameter specifically affect the AUROC(%) and ranking of the CAE model. It can be seen that

Table B12 AUROC(%)/Ranking of the DAE on the validation set

Abnomal class	**Latent Dimension** 32	64	128	256	512	1024
plane	39.8/9	43.85/5	41.3/7	45.81/4	49.34/1	48.33/2
car	73.39/5	74.08/1	73.79/2	73.45/4	73.16/6	72.76/9
bird	39.62/3	36.78/12	37.78/6	38.49/4	40.88/2	41.3/1
cat	54.41/8	54.27/9	53.91/10	53.01/13	53.65/12	53.73/11
deer	33.43/8	34.55/5	35.37/4	36.98/3	38.03/2	40/1
dog	56.2/8	56.15/9	55.64/10	55.33/11	54.27/12	53.98/13
frog	47.81/6	53.82/5	54.15/4	57.54/3	58.65/2	59.84/1
horse	61.87/6	61.87/6	61.22/8	60.85/10	60.9/9	60.69/11
ship	40.34/10	40.13/12	40.95/7	39.75/13	41.97/4	40.3/11
truck	74.18/4	73.74/7	73.74/7	73.34/9	72.68/11	71.87/12
Average	6.7	7.1	6.5	7.4	6.1	7.2

Table B13 Architecture and hyperparameters of the DAE

Operation	**Units**	**Non Linearity**	**Dropout**
Encoder			
Dense	1024	Tanh	0.0
Dense	256	Tanh	0.0
Dense	12	Tanh	0.0
Decoder			
Dense	12	Tanh	0.0
Dense	256	Tanh	0.0
Dense	1024	Tanh	0.0
Optimizer	Adam		
Batch size	64		
Latent dimension	12		
Epochs	40		
Weight initialization	Gaussian(μ=0, σ=0.02)		
Bias initialization	Constant(0)		

the optimal ranking of the CAE is achieved when the dimension of the latent representation is 512 for the gas turbine data set, which is shown in Table B16.

B.3 Architecture and hyperparameters of the CSiamese

Table B17 and Table B18 depict how the choice of the hyperparameter specifically affect the AUROC(%) and ranking of the CSiamese model. It can be seen that the optimal ranking of the CSiamese is achieved when the dimension of the latent representation is 6 for the gas turbine data set, which is shown in Table B19.

Table B14 AUROC(%)/Ranking of the CAE on the validation setExample of a lengthy table which is set to full textwidth

Abnomal Class	Latent Dimension 2	4	6	8	10	12	16
plane	67.03/12	64.11/13	71.4/10	71.17/11	75.5/9	76.19/8	78.5/7
car	70.11/13	73.36/12	75.05/10	75.1/9	75/11	76.3/8	76.6/7
bird	37.99/13	39.15/12	40.51/11	41.23/10	41.86/8	41.78/9	42.22/7
cat	58.25/13	60.24/6	58.33/12	59.14/8	58.47/11	59.8/7	59.01/9
deer	38.14/13	42.58/12	42.62/11	45.96/10	46.14/9	49.11/8	51.38/7
dog	62.85/1	62.26/3	62.08/4	60.48/9	61.52/5	60.73/8	61.12/7
frog	47.45/13	54.22/11	53.33/12	56.94/10	60.56/9	64.26/7	64.22/8
horse	59.32/13	62.92/12	63.23/11	63.83/10	65.51/5	65.33/6	65.01/7
ship	55.58/13	56.22/12	65.32/6	59.26/10	60.53/9	58.35/11	62.12/8
truck	71.37/13	73.72/12	76.19/9	75.79/10	75.48/11	77.46/7	77.19/8
Average	11.7	10.5	9.6	9.7	8.7	7.9	7.5

Table B15 AUROC(%)/Ranking of the CAE on the validation set

Abnomal class	Latent Dimension 32	64	128	256	512	1024
plane	84.6/6	92.46/5	93.47/4	95.93/3	96.65/2	98.49/1
car	79.09/6	81.14/5	84.9/4	87.91/2	88.16/1	87.73/3
bird	43.85/6	45.06/5	49.45/4	51.03/3	58.58/2	63.16/1
cat	62.04/4	58.65/10	60.38/5	64.67/3	66.84/1	65.29/2
deer	56.07/6	63.41/5	65.46/4	69.13/3	74.48/2	76.83/1
dog	59.34/10	59.19/11	56.91/13	58.91/12	62.63/2	61.19/6
frog	68.69/6	78.21/5	81.56/4	87.41/3	90.68/2	92.15/1
horse	63.87/9	64.16/8	66.76/4	67.14/3	67.67/2	69.71/1
ship	64.95/7	73.13/5	79.19/4	84.16/3	87.14/2	90.36/1
truck	77.75/6	78.88/5	82.33/3	80.67/4	86.65/1	84.92/2
Average Ranking	6.6	6.4	4.9	3.9	**1.7**	1.9

Table B16 Architecture and hyperparameters of the CAE

Operation	Kernel	Strides	Filters Units	Non Linearity	Batch Norm
Encoder					
Conv	4×4	2×2	128	LeakyReLU(0.2)	✓
Conv	4×4	2×2	256	LeakyReLU(0.2)	✓
Conv	4×4	2×2	512	LeakyReLU(0.2)	✓
Conv	4×4	2×2	512	None	×
Decoder					
DeConv	4×4	2×2	512	ReLU	✓
DeConv	4×4	2×2	512	ReLU	✓
DeConv	4×4	2×2	256	ReLU	✓
DeConv	4×4	2×2	128	Tanh	×

Optimizer	Adam
Batch size	64
Latent dimension	512
Epochs	15
Weight initialization	Gaussian(μ=0, σ=0.02)
Bias initialization	Constant(0)

Table B17 AUROC(%)/Ranking of the CSiamese on the validation set

Abnomal class	**Latent Dimension** 2	4	6	8	10	12	16
plane	93.68/11	98.53/1	98.37/2	96.31/7	97.21/4	97.88/3	97.02/5
car	58.73/10	60.33/9	70.37/2	70.69/1	69.61/3	65.07/5	66.31/4
bird	57.86/8	61.88/6	68.69/1	65.49/3	65.46/4	66.07/2	65.03/5
cat	70.15/6	73.47/1	71.44/2	71.06/5	71.27/4	71.39/3	69.17/7
deer	65.77/13	89.75/2	84.96/5	89.99/1	88.23/3	84.92/6	86.15/4
dog	85.82/1	80.86/2	80.08/3	75.67/5	77.09/4	74.48/7	75.02/6
frog	77.34/11	91.92/6	93.4/3	94.2/1	93.28/4	93.89/2	92.12/5
horse	66.7/3	72.58/2	74.07/1	62.07/7	63.58/6	63.99/5	64.16/4
ship	65.3/13	94.89/2	95.76/1	94.09/4	94.18/3	90.69/7	91.48/5
truck	61.49/9	61.39/10	72.33/3	57.53/12	71.21/4	74.52/2	77.28/1
Average	8.5	4.1	**2.3**	4.6	3.9	4.2	4.6

Table B18 AUROC(%)/Ranking of the CSiamese on the validation set

Abnomal class	Latent Dimension 32	64	128	256	512	1024
plane	96.46/6	95.9/8	95.83/9	92.98/13	93.91/10	93.43/12
car	64.9/6	62.15/7	60.34/8	56.16/13	56.83/12	58.04/11
bird	57.88/7	57.32/9	56.57/11	55.46/13	55.71/12	57.14/10
cat	67.56/8	63.99/9	60.1/13	60.14/12	61.78/11	62.51/10
deer	79.74/7	77.13/8	67.06/12	69.35/11	70.65/9	70.46/10
dog	70.41/8	68.74/9	64.77/10	62.2/11	60.27/12	58.77/13
frog	90.75/7	86.57/8	76.29/13	76.75/12	79.42/10	84.6/9
horse	60.14/9	60.93/8	57.38/13	58.28/11	59.92/10	57.83/12
ship	91.21/6	85.86/8	83.73/9	81.77/11	76.68/12	83.62/10
truck	65.25/6	63.67/7	67.08/5	62.23/8	59.81/11	57.23/13
Average	7	8.1	10.3	11.5	10.9	11

Table B19 Architecture and hyperparameters of the CSiamese

Operation	Kernel	Strides	Filters Units	Non Linearity	Batch Norm
Encoder					
Conv	4×4	2×2	128	LeakyReLU(0.2)	✓
Conv	4×4	2×2	256	LeakyReLU(0.2)	✓
Conv	4×4	2×2	512	LeakyReLU(0.2)	✓
Conv	4×4	2×2	6	None	×
Decoder					
DeConv	4×4	2×2	6	ReLU	✓
DeConv	4×4	2×2	512	ReLU	✓
DeConv	4×4	2×2	256	ReLU	✓
DeConv	4×4	2×2	128	Tanh	×
Siamese					
Conv	4×4	2×2	128	LeakyReLU(0.2)	✓
Conv	4×4	2×2	256	LeakyReLU(0.2)	✓
Conv	4×4	2×2	512	LeakyReLU(0.2)	✓
Conv	4×4	2×2	6	None	×
Optimizer	Adam				
Batch size	64				
Latent dimension	6				
Epochs	(5, 10)				
Weight initialization	Gaussian(μ=0, σ=0.02)				
Bias initialization	Constant(0)				